

Appendix: Mixing Non-interference with Noncompliance

The voting example in section 8.1 is relatively tame by interference standards. Spillovers are confined to households, and identification of causal effects is straightforward given the multi-level experimental design. Even when spillovers are narrowly confined, identification becomes much more complex when experiments involve noncompliance. Thinking back to Chapters 5 and 6, recall that noncompliance refers to situations in which the experimental assignments do not match the treatments that subjects actually receive. Noncompliance forces us to grapple with the problem of heterogeneous treatment effects among different latent types of people: Compliers, Never-Takers, Always-Takers, and Defiers. Mixing noncompliance and interference expands the schedule of potential outcomes to allow for direct and spillover effects among each latent type. Even relatively simple instances where spillovers mix with noncompliance force the researcher to make additional assumptions in order to identify causal effects.

In order to practice setting up a model to identify causal parameters, let's see how one-sided noncompliance plays out in the previous example. Suppose that instead of mailing voters, the campaign sought to speak to them by phone in an effort to encourage voter turnout. Because only those slated for treatment can receive the campaign's call, there are no Always-Takers or Defiers. (Always-Takers and Defiers would take the treatment if assigned to the control group, but cannot because no one in the control group is ever called.) Each two-voter household therefore contains some combination of Compliers (who answer if called) and Never-Takers (who do not answer if called). There are four possible types of households: both voters are Compliers, both are Never-Takers,

only the targeted voter is a Complier, and only the non-targeted voter is a Complier. When we randomly assign two-person households to receive 0, 1, or 2 calls based on the same assignment schedule as in the mail experiment above, we can collect the following pieces of information. First, we obtain voter turnout rates for each person in the household. Second, we know whether we were able to contact each targeted person. The voting data enable us to compare voting rates across the four experimental groups: those where both voters are targeted, those where neither voter is targeted, those where a given voter is targeted but the housemate is not, and those where a given voter is not targeted but the housemate is. The contact data enable us to estimate the proportions of households with two Compliers, one Complier, and no Compliers.

This seems like a promising start, but the difficulties become apparent when we write out the expected voting rate in each of the randomly assigned groups. Each type of household has an unknown expected rate of voting in the absence of treatment:

$\omega_{NN}, \omega_{CN}, \omega_{NC}, \omega_{CC}$, where the subscripts stand for whether the subject's housemate and the subject are Never-Takers or Compliers. We do not know whether these expected rates vary from one type of household to the next, and we would like to develop a model that is agnostic on this point. We also want to remain agnostic as to how many households of each type there are in our sample. The proportion of voters of each type may be written $\alpha_{NN}, \alpha_{CN}, \alpha_{NC}, \alpha_{CC}$. Because these proportions must sum to 1, we require that $\alpha_{NN} = 1 - \alpha_{CN} - \alpha_{NC} - \alpha_{CC}$.

Let $\bar{Y}_{00}, \bar{Y}_{10}, \bar{Y}_{01}, \bar{Y}_{11}$ represent the expected voting rates in each randomly assigned experimental group, where the first subscript refers to whether the subject's housemate is targeted for treatment, and the second subscript refers to whether the subject is targeted.

(Note that random assignment determines the groupings; we do not partition voters according to who is actually contacted for reasons explained in Chapters 5 and 6.) The expected voting rate in each randomly assigned group is a weighted average of the baseline voting propensities, the direct treatment effects, and the spillover effects. Random assignment ensures that each experimental group contains the same expected proportions of each voter type. In other words, we expect each experimental group to contain α_{NN} share of households with two Never-Takers, α_{CC} share of households with two Compliers, and so forth.

The expected voting rate of voters in households assigned to receive no calls is:

$$\bar{Y}_{00} = \alpha_{CC}\omega_{CC} + \alpha_{NC}\omega_{NC} + \alpha_{CN}\omega_{CN} + (1 - \alpha_{CN} - \alpha_{NC} - \alpha_{CC})\omega_{NN}. \quad (\text{A.1})$$

This formula is less complicated than it first appears. Every group's base rate of voting is multiplied by the share of all voters belonging to that group, producing a weighted average of the four groups' base rates.

In order to see how this weighted average changes when a household is assigned to treatment, consider the expected voting rate for people who are targeted for a call but whose housemates are not:

$$\bar{Y}_{01} = \alpha_{CC}(\omega_{CC} + t'_D) + \alpha_{NC}(\omega_{NC} + t_D) + \alpha_{CN}\omega_{CN} + (1 - \alpha_{CN} - \alpha_{NC} - \alpha_{CC})\omega_{NN}. \quad (\text{A.2})$$

Notice that the base rates of voting among targeted Compliers have been augmented by a treatment effect. In households where the targeted voter is the only Complier, the

average treatment effect is t_D ; in households with two Compliers, the average treatment effect is t'_D .

As we turn to people who receive no call themselves but whose housemates are called, we see how expected voting rates are affected by spillovers:

$$\bar{Y}_{10} = \alpha_{CC}(\omega_{CC} + t'_S) + \alpha_{NC}\omega_{NC} + \alpha_{CN}(\omega_{CN} + t_S) + (1 - \alpha_{CN} - \alpha_{NC} - \alpha_{CC})\omega_{NN}. \quad (\text{A.3})$$

Again, we allow for different spillover effects among those households with one targeted Complier and those with two Compliers. For those in households with one Complier, the treatment effect is t_S ; in households with two Compliers, the treatment effect is t'_S .

Finally, we come to voters in households where both members are called. The expected vote is a blend of the earlier equations, with one addition:

$$\bar{Y}_{11} = \alpha_{CC}(\omega_{CC} + t'_D + t'_S + \Delta) + \alpha_{NC}(\omega_{NC} + t_D) + \alpha_{CN}(\omega_{CN} + t_S) + (1 - \alpha_{CN} - \alpha_{NC} - \alpha_{CC})\omega_{NN}. \quad (\text{A.4})$$

The treatment effect parameter Δ acknowledges the possibility that direct and spillover effects may interact – the spillover effect on a Complier who has received the treatment may be different from the spillover effect on a Complier who was not called.

Although our experiment allows us to directly estimate α_{CN} , α_{NC} , α_{CC} , and ω_{NN} , we cannot solve for the five treatment parameters with just four equations. We're stuck unless we impose some additional assumptions. What happens if we assume that $t_D = t'_D$ and $t_S = t'_S$? The identification problem becomes much easier. To identify the direct effect, we can subtract equation (A.1) from equation (A.2) to obtain:

$$\bar{Y}_{01} - \bar{Y}_{00} = (\alpha_{CC} + \alpha_{NC})t_D. \quad (\text{A.5})$$

In order to estimate t_D , we use our sample estimates. We have an estimate of the voting rates among individuals in households that received no calls ($\hat{Y}_{00}$) and among those who were targeted themselves but their housemates were not ($\hat{Y}_{01}$). Based on information gathered when the phone bank attempted to reach targeted voters, we have information about the proportion of households in which just one person is reachable and in which both people are reachable. Since our experimental groups are of equal size, the estimator of the direct treatment effect is:

$$\hat{t}_D = \frac{\hat{Y}_{01} - \hat{Y}_{00}}{\hat{\alpha}_{NC} + \hat{\alpha}_{CC}}. \quad (\text{A.6})$$

Among households with at least one targeted Complier, $\hat{t}_D$ represents the estimated average effect of receiving a call directly.

The same procedure allows us to identify and estimate the spillover effect among those whose housemates are Compliers. The difference in expected voting rates between those who are not targeted themselves but reside with a targeted housemate ($\bar{Y}_{10}$) and those who are in untargeted households ($\bar{Y}_{00}$) may be written:

$$\bar{Y}_{10} - \bar{Y}_{00} = (\alpha_{CC} + \alpha_{CN})t_S, \quad (\text{A.7})$$

which suggests a useful estimator:

$$\hat{t}_S = \frac{\hat{Y}_{10} - \hat{Y}_{00}}{\hat{\alpha}_{CN} + \hat{\alpha}_{CC}}. \quad (\text{A.8})$$

We are able to identify both the direct and spillover effects without using the group in which both household members are called, which means that this group is in some sense superfluous. This group's contribution is the identification of the interaction effect, Δ , which indicates the extent to which the effects (direct or spillover) change for households in which both voters receive a call. The expected difference-in-differences yields:

$$\bar{Y}_{11} - \bar{Y}_{10} - (\bar{Y}_{01} - \bar{Y}_{00}) = \alpha_{CC}\Delta, \quad (\text{A.9})$$

which means that the interaction can be identified by:

$$\Delta = \frac{\bar{Y}_{11} - \bar{Y}_{10} - (\bar{Y}_{01} - \bar{Y}_{00})}{\alpha_{CC}}. \quad (\text{A.10})$$

In order to estimate this interaction, simply replace the expected voting rates with the observed voting rates and the estimate of α_{CC} .

This type of exercise is typical of what one encounters when analyzing spillover using experiments that involve noncompliance. Noncompliance forces the researcher to come to terms with the possibility that latent groups might respond differently to the treatment. Spillovers, meanwhile, expand the number of potential outcomes with which the researcher must contend. When the two are mixed together, identification requires that the analyst rein in some of the heterogeneity in treatment and spillover effects.

Whether the constraints that we imposed above make sense substantively will depend on the empirical application. In this particular example, intuition might suggest that individuals residing in two-Complier households and individuals residing in one-Complier households respond similarly to the treatment – but intuitions can be mistaken.

When confronting both noncompliance and spillover, researchers may avoid some of the headaches of this section by implementing a placebo design (see Chapter 5). A placebo design in this case means that calls are placed on irrelevant topics (e.g., encouraging people to fasten their safety belts when driving) simply to find out who answers the phone. By gauging who is a Complier or Never-Taker, the placebo creates control groups that parallel each of the experimental treatment groups. This design allows for a direct comparison between, for example, voters in two-Complier households where both voters receive a placebo call and voters who receive a placebo call while their housemate receives a treatment call. Although challenging to orchestrate, this type of design allows the researcher to sidestep some of the identification problems that arise due to noncompliance. When rates of noncompliance are high, a placebo design may also generate much more precise estimates than a standard design involving a treatment group and an untreated control group.